

## SAFETY DATA SHEET

Creation Date 25-Mar-2010

Revision Date 18-Jan-2018

Revision Number 3

### 1. Identification

**Product Name** SurePrep™ RNA/DNA/Protein Purification Kit, Protein Neutralizer

**Cat No. :** BP2802-4; BP2802-50; BP2802-4J; BP2802-50J

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Not for food, drug, pesticide or biocidal product use

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

##### **Emergency Telephone Number**

CHEMTREC®, Inside the USA: 800-424-9300

CHEMTREC®, Outside the USA: 001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                      |            |
|------------------------------------------------------|------------|
| Corrosive to metals                                  | Category 1 |
| Skin Corrosion/Irritation                            | Category 1 |
| Serious Eye Damage/Eye Irritation                    | Category 1 |
| Specific target organ toxicity (single exposure)     | Category 3 |
| Target Organs - Respiratory system.                  |            |
| Specific target organ toxicity - (repeated exposure) | Category 2 |
| Target Organs - Liver, Blood.                        |            |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May be corrosive to metals

Causes severe skin burns and eye damage

May cause respiratory irritation

May cause damage to organs through prolonged or repeated exposure



### Precautionary Statements

#### Prevention

Do not get in eyes, on skin, or on clothing  
Do not breathe dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Wash face, hands and any exposed skin thoroughly after handling

#### Response

Get medical attention/advice if you feel unwell

#### Inhalation

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Call a POISON CENTER or doctor/physician if you feel unwell

#### Skin

Immediately call a POISON CENTER or doctor/physician  
Wash contaminated clothing before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

#### Eyes

Immediately call a POISON CENTER or doctor/physician  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

#### Ingestion

IF SWALLOWED: Rinse mouth. DO NOT induce vomiting

#### Storage

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

#### Disposal

Dispose of contents/container to an approved waste disposal plant

#### Hazards not otherwise classified (HNOC)

None identified

## 3. Composition/Information on Ingredients

| Component            | CAS-No    | Weight % |
|----------------------|-----------|----------|
| Nuclease Free Water  | 7732-18-5 | 98 - 100 |
| Citric acid          | 77-92-9   | 1 - 3    |
| Orthophosphoric acid | 7664-38-2 | < 1      |

## 4. First-aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Inhalation

Move to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

|                                            |                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------|
| <b>Ingestion</b>                           | Do not induce vomiting. Call a physician or Poison Control Center immediately. |
| <b>Most important symptoms and effects</b> | No information available.                                                      |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                          |

## 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Substance is nonflammable; use agent most appropriate to extinguish surrounding fire. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | Not applicable                                                                        |
| <b>Method -</b>                         | No information available                                                              |
| <b>Autoignition Temperature</b>         | No information available                                                              |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | No data available                                                                     |
| <b>Lower</b>                            | No data available                                                                     |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Carbon monoxide (CO) Carbon dioxide (CO<sub>2</sub>) Oxides of phosphorus

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional ecological information.           |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.                     |

## 7. Handling and storage

|                 |                                                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Use only under a chemical fume hood. Wear personal protective equipment. Do not breathe vapors or spray mist. Do not get in eyes, on skin, or on clothing. |
| <b>Storage</b>  | Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.                                                                  |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component            | ACGIH TLV                                             | OSHA PEL                                                                                              | NIOSH IDLH                                                                            | Mexico OEL (TWA)                                      |
|----------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------|
| Orthophosphoric acid | TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup><br>(Vacated) STEL: 3 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | IDLH: 1000 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: The National Institute for Occupational Safety and Health Immediately Dangerous to Life or Health

**Engineering Measures** Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

**Personal Protective Equipment**

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Liquid                   |
| Appearance                             | Clear Colorless          |
| Odor                                   | No information available |
| Odor Threshold                         | No information available |
| pH                                     | 1.5                      |
| Melting Point/Range                    | No data available        |
| Boiling Point/Range                    | No information available |
| Flash Point                            | Not applicable           |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

**Reactive Hazard** None known, based on information available

**Stability** Stable under normal conditions.

**Conditions to Avoid** Incompatible products. Excess heat.

**Incompatible Materials** Strong oxidizing agents

**Hazardous Decomposition Products** Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Oxides of phosphorus

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

### Component Information

| Component            | LD50 Oral                                          | LD50 Dermal                  | LC50 Inhalation                   |
|----------------------|----------------------------------------------------|------------------------------|-----------------------------------|
| Nuclease Free Water  | LD50 > 90 mL/kg ( Rat )                            | Not listed                   | Not listed                        |
| Citric acid          | LD50 = 3 g/kg ( Rat )<br>LD50 = 3000 mg/kg ( Rat ) | >2 g/kg ( Rat )              | Not listed                        |
| Orthophosphoric acid | 2600 mg/kg (Rat)                                   | LD50 = 2740 mg/kg ( Rabbit ) | 850 mg/m <sup>3</sup> ( Rat ) 1 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Causes burns by all exposure routes

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component            | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------|-----------|------------|------------|------------|------------|------------|
| Nuclease Free Water  | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Citric acid          | 77-92-9   | Not listed | Not listed | Not listed | Not listed | Not listed |
| Orthophosphoric acid | 7664-38-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** Liver Blood

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated. See actual entry in RTECS for complete information.

## 12. Ecological information

### Ecotoxicity

| Component            | Freshwater Algae | Freshwater Fish                         | Microtox                                          | Water Flea             |
|----------------------|------------------|-----------------------------------------|---------------------------------------------------|------------------------|
| Citric acid          | Not listed       | Leuciscus idus: LC50 = 440-760 mg/L/96h | Photobacterium phosphoreum: EC50 = 14 mg/L/15 min | EC50 = 120 mg/L/72h    |
| Orthophosphoric acid | Not listed       | 98 - 106 mg/L LC50 96 h                 | Not listed                                        | > 100 mg/L EC50 = 48 h |

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility**

| Component   | log Pow |
|-------------|---------|
| Citric acid | -1.72   |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN1805  
 Proper Shipping Name PHOSPHORIC ACID SOLUTION  
 Hazard Class 8  
 Packing Group III

### TDG

UN-No UN1805  
 Proper Shipping Name PHOSPHORIC ACID SOLUTION  
 Hazard Class 8  
 Packing Group III

### IATA

UN-No UN1805  
 Proper Shipping Name PHOSPHORIC ACID SOLUTION  
 Hazard Class 8  
 Packing Group III

### IMDG/IMO

UN-No UN1805  
 Proper Shipping Name PHOSPHORIC ACID SOLUTION  
 Hazard Class 8  
 Packing Group III

## 15. Regulatory information

### International Inventories

| Component            | TSCA | DSL | NDSL | EINECS    | ELINCS | NLP | PICCS | ENCS | AICS | IECSC | KECL |
|----------------------|------|-----|------|-----------|--------|-----|-------|------|------|-------|------|
| Nuclease Free Water  | X    | X   | -    | 231-791-2 | -      |     | X     | -    | X    | X     | X    |
| Citric acid          | X    | X   | -    | 201-069-1 | -      |     | X     | X    | X    | X     | X    |
| Orthophosphoric acid | X    | X   | -    | 231-633-2 | -      |     | X     | X    | X    | X     | X    |

#### Legend:

X - Listed

E - Indicates a substance that is the subject of a Section 5(e) Consent order under TSCA.

F - Indicates a substance that is the subject of a Section 5(f) Rule under TSCA.

N - Indicates a polymeric substance containing no free-radical initiator in its inventory name but is considered to cover the designated polymer made with any free-radical initiator regardless of the amount used.

P - Indicates a commenced PMN substance

R - Indicates a substance that is the subject of a Section 6 risk management rule under TSCA.

S - Indicates a substance that is identified in a proposed or final Significant New Use Rule

T - Indicates a substance that is the subject of a Section 4 test rule under TSCA.

XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B)).

Y1 - Indicates an exempt polymer that has a number-average molecular weight of 1,000 or greater.

Y2 - Indicates an exempt polymer that is a polyester and is made only from reactants included in a specified list of low concern reactants that comprises one of the eligibility criteria for the exemption rule.

### U.S. Federal Regulations

**TSCA 12(b)** Not applicable

**SARA 313** Not applicable

**SARA 311/312 Hazard Categories** See section 2 for more information

### **CWA (Clean Water Act)**

| Component            | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|----------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Orthophosphoric acid | X                          | 5000 lb                     | -                      | -                         |

**Clean Air Act** Not applicable

**OSHA Occupational Safety and Health Administration**  
Not applicable

**CERCLA** This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

| Component            | Hazardous Substances RQs | CERCLA EHS RQs |
|----------------------|--------------------------|----------------|
| Orthophosphoric acid | 5000 lb                  | -              |

**California Proposition 65** This product does not contain any Proposition 65 chemicals

### **U.S. State Right-to-Know Regulations**

| Component            | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|----------------------|---------------|------------|--------------|----------|--------------|
| Nuclease Free Water  | -             | -          | X            | -        | -            |
| Orthophosphoric acid | X             | X          | X            | -        | X            |

### **U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

### **U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

### Other International Regulations

**Mexico - Grade** No information available

## **16. Other information**

**Prepared By** Regulatory Affairs  
Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

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|                         |                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Creation Date</b>    | 25-Mar-2010                                                                                                                                                                                                                             |
| <b>Revision Date</b>    | 18-Jan-2018                                                                                                                                                                                                                             |
| <b>Print Date</b>       | 18-Jan-2018                                                                                                                                                                                                                             |
| <b>Revision Summary</b> | This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS). |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**